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Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India.

DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

SKILL BASED EVALUATION REPORT

Academic Year 2026-2027

ODD SEMESTER

Title of the Microproject : College Club Management System (CCMS)

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Register Number : URK2CS1016, URK25CS1

Course Code : 25CS312

Course Name : Software Engineering

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Abstract

College clubs are the heartbeat of campus life, yet students discover and participate in them through scattered Google Forms, QR codes, WhatsApp groups, posters and separate registration links. Information about memberships, events, announcements, attendance and participation history remains fragmented across multiple platforms, creating confusion for students and heavy manual effort for club coordinators. This micro project presents the design and development of the **College Club Management System (CCMS)**, a centralized web application that brings the entire club ecosystem onto a single platform. Students can discover all college clubs, filter them by category, view rich club profiles, apply for membership, register for events, receive announcements, and track attendance and participation history from one dashboard. Club coordinators can manage club information, review membership applications, create events, manage registrations, mark attendance, publish announcements and view analytics. College administrators can approve clubs, manage users and monitor institution-level activity. CCMS is built using React, TypeScript and Vite for the front end, styled with Tailwind CSS, and powered by Supabase (PostgreSQL, Authentication and Row Level Security) for the back end. The system was developed following an iterative Agile software engineering process with UML-based design and systematic unit testing. The results demonstrate a secure, responsive and user-friendly platform that eliminates manual coordination and makes club activities accessible to every student.

Keywords: Club Management System, Web Application, React, Supabase, Software Engineering, Student Engagement.

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Chapter 1: Introduction

1.1 Background of the Project

College clubs - technical, cultural, literary and sports - play a vital role in the overall development of students. They provide opportunities to build technical skills, develop leadership qualities, showcase talent and work in teams. In most colleges, however, the management of these clubs is highly fragmented. Students learn about clubs through posters, notice boards, forwarded WhatsApp messages, QR codes and Google Forms. Each club maintains its own registration process, event notices and attendance sheets.

With the increasing number of clubs and activities every semester, manual coordination becomes error-prone and time-consuming. Important updates are lost in chat groups, attendance records are maintained in spreadsheets and students have no single place to view their participation history. A dedicated web-based system can solve these problems by unifying club discovery, membership management, event registration, attendance tracking and announcements on one platform. This micro project, the College Club Management System (CCMS), is designed and developed as part of the Software Engineering course (25CS312) applying the complete software development life cycle.

1.2 Problem Statement

Students often discover clubs through different channels - Google Forms sent through email, QR codes linked to forms, WhatsApp messages, posters and separate registration links. Even after joining a club, students struggle to find event information, announcements, attendance records and participation history because these are spread across multiple platforms. Club coordinators manage membership lists and attendance manually using spreadsheets, which is tedious and error-prone. College administrators have no centralized view of club activities across the institution. Therefore, there is a clear need for a centralized College Club Management System through which every student can discover, join and participate in clubs, coordinators can manage their clubs efficiently, and administrators can monitor the complete club ecosystem of the college.

1.3 Objectives

- To develop a single centralized web platform that lists all academic and non-academic clubs of the college.
- To enable students to search and filter clubs, view club profiles and apply for membership online.
- To provide an event hub where students can view upcoming events and register with a single tap.
- To automate attendance marking and maintain a complete participation history for every student.
- To deliver instant announcements from clubs to members through a unified feed.
- To provide role-based dashboards for students, club coordinators and administrators.
- To generate club-level and institution-level analytics and reports.

- To ensure secure access to data using authentication and row-level security policies.

1.4 Scope of the Project

The system covers three user roles: **Student**, **Club Coordinator** and **Administrator**. The student module provides club discovery, membership application, event registration, announcements, attendance history and an activity portfolio. The coordinator module provides membership approval, event creation and management, attendance marking, announcements and club analytics. The administrator module provides club approval, user and coordinator management, category management and institution-level analytics. The front end is built with React and TypeScript, the back end services and database are provided by Supabase, and the application is deployed on the web. Features such as online payments for paid events, mobile applications and AI-based recommendations are considered out of scope for this micro project and are identified as future enhancements.

Chapter 2: Literature Survey

2.1 Review of Existing Research or Related Projects

The existing practice followed in most colleges is a combination of Google Forms, Google Sheets and WhatsApp groups. Google Forms provide quick data collection and are widely used for event registration, while Sheets are used to store membership and attendance data [9]. Although these tools are easy to use, they operate in isolation: every club creates its own form and sheet, students must fill multiple forms for multiple clubs and coordinators manually reconcile the collected data.

Commercial event and campus engagement platforms such as Eventbrite [8] and similar community management products provide polished event creation, ticketing and RSVP features. However, these platforms are designed for public events, are generally paid for advanced features, and do not model college-specific entities such as club categories, faculty coordinators, semester-wise participation or academic attendance records.

Several open-source web-based club and student organization management systems reported in the literature implement CRUD workflows for members, events and notices using PHP/MySQL or Java stacks [7]. Recent work on student engagement platforms highlights that a centralized information system improves participation rates and reduces administrative overhead. Modern web engineering practice recommends component-based front ends with managed back-end services (Backend-as-a-Service) for rapid, secure development [1][3][4].

2.2 Summary of Existing Approaches

Approach	Strengths	Limitations
Manual (posters, word of mouth)	No technology needed	Very low reach; information is easily missed
Google Forms + WhatsApp	Quick to set up; familiar to students	Fragmented data; duplicate entries; no unified history
Spreadsheet registers	Flexible; free	Manual effort; error-prone; single-owner bottleneck
Generic event platforms	Polished event pages; RSVP features	Not college-specific; paid; no attendance or portfolio
CCMS (proposed system)	Single platform; role-based; secure; analytics	Requires internet access and initial data setup

Table 2.1: Comparison of Existing Approaches

2.3 Research Gap or Limitations of Existing Systems

- No single system integrates club discovery, membership, events, attendance and announcements for a college.
- Existing tools do not provide role-based dashboards for students, coordinators and administrators together.
- Participation history and activity portfolios of students are not maintained in a shareable digital form.

- Manual attendance and spreadsheet-based registers cannot provide real-time analytics to administrators.
- Generic platforms do not enforce college-specific access control and data privacy requirements.

Chapter 3: Methodology and System Design

3.1 Proposed Methodology

CCMS is developed using an **iterative Agile software development process**. In each iteration the team performs requirement gathering, design, implementation, testing and review. Requirements were collected by analysing the existing club registration workflow of the college and by studying the needs of students, club coordinators and administrators. The system design follows unified modelling language (UML) practices: use case diagrams capture the functional requirements of each actor, and an entity-relationship diagram models the persistent data. Implementation follows component-based development with clear separation of concerns - user interface (React components), business logic (service modules) and data (Supabase PostgreSQL) - achieving low coupling and high cohesion [1][2]. Testing is carried out at unit and integration levels, and the application is version-controlled using Git and GitHub with deployment on the Vercel platform.

3.2 System Architecture or Block Diagram

The system follows a three-tier architecture as shown in Figure 3.1. The **client tier** is a single-page application built with React and TypeScript, rendered in the browser, with separate views for students, coordinators and administrators. The **application tier** consists of the Supabase platform which exposes a RESTful API generated from the database schema, handles authentication using JSON Web Tokens and enforces Row Level Security (RLS) policies. The **data tier** is a PostgreSQL database that stores profiles, clubs, memberships, events, registrations, attendance and announcements.

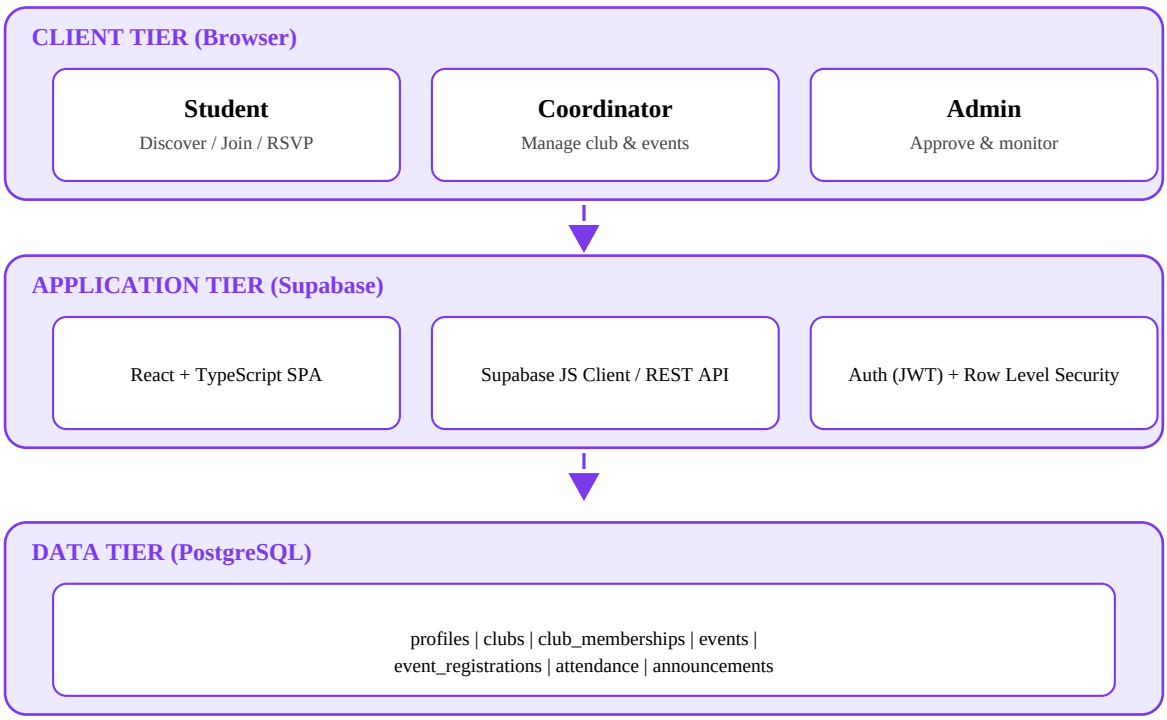


Figure 3.1: System Architecture of CCMS

3.3 UML Diagrams

The functional model of the system is captured using the use case diagram shown in Figure 3.2. The three actors - Student, Club Coordinator and Admin - interact with the system through well-defined use cases.

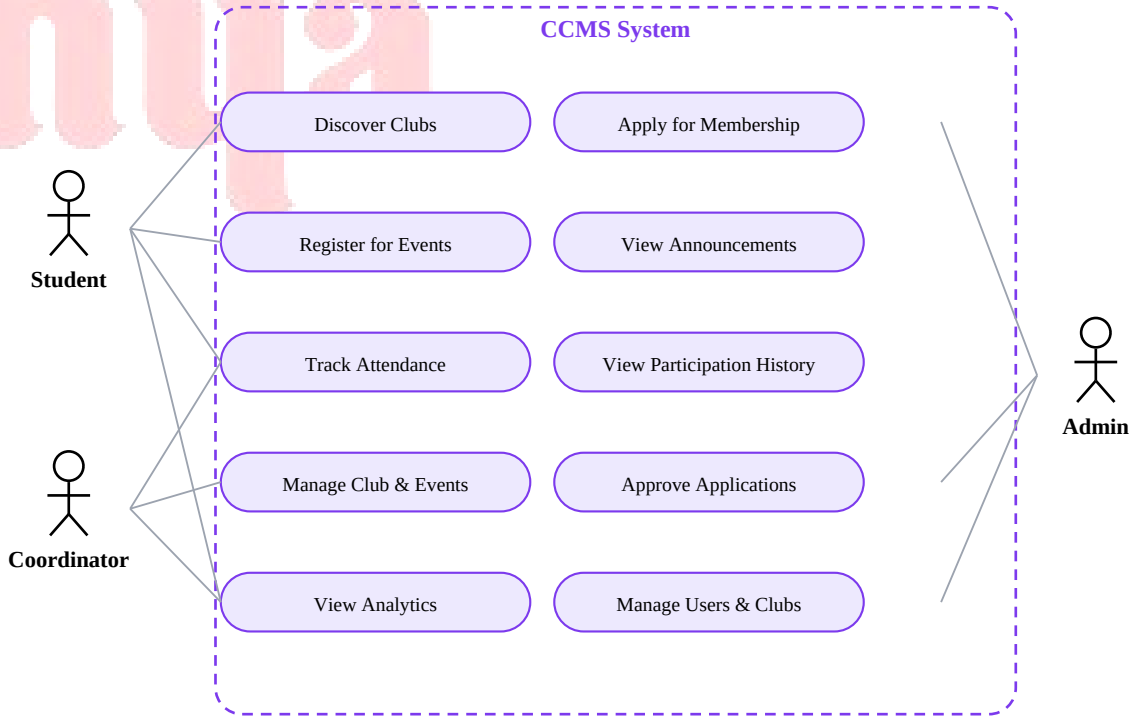


Figure 3.2: Use Case Diagram of CCMS

The data model is designed using the entity-relationship diagram shown in Figure 3.3. The main entities are profiles, club_categories, clubs, club_memberships, events, event_registrations, attendance and announcements, with one-to-many and many-to-one relationships between them.

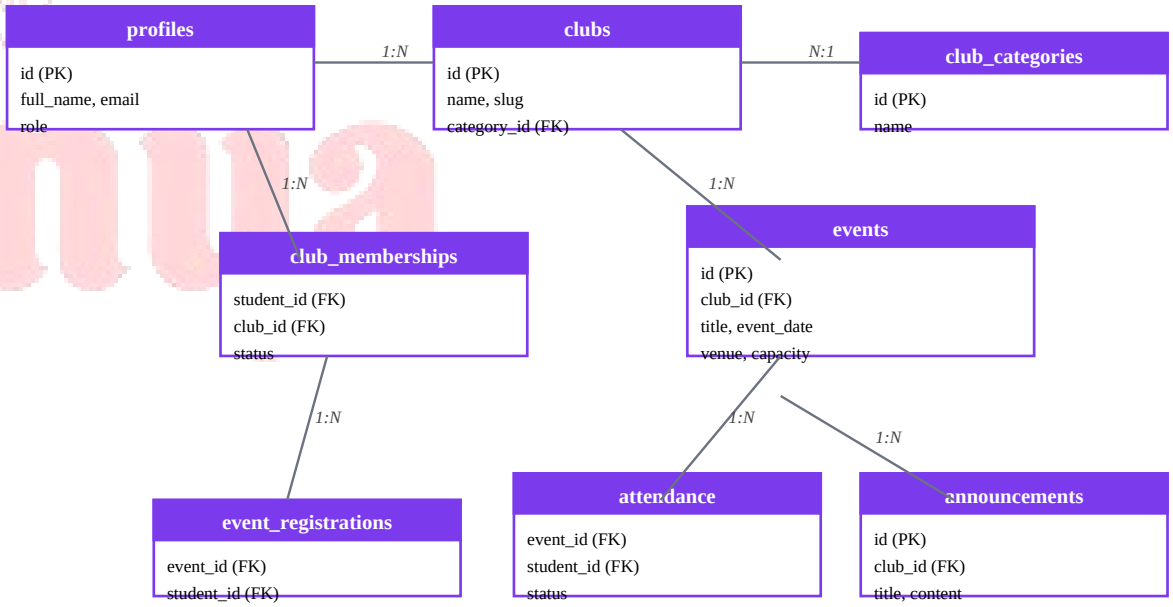


Figure 3.3: Entity-Relationship Diagram of CCMS

The functional decomposition of the system into modules is shown in Figure 3.4. Each module groups the features available to a specific user role.

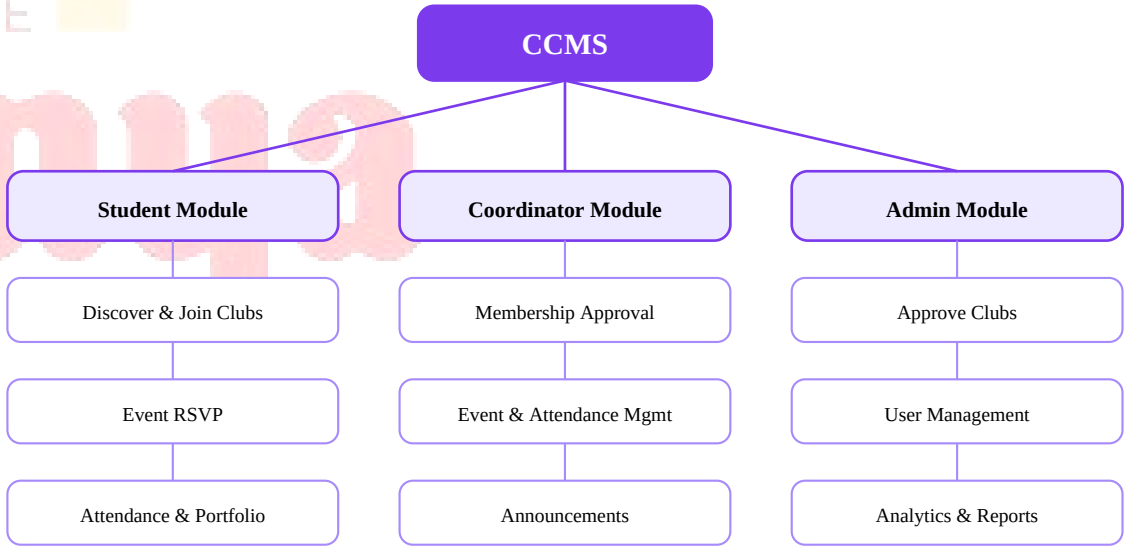


Figure 3.4: Module Structure of CCMS

3.4 Tools and Technologies Used

Category	Tool / Technology	Purpose
Front-end library	React 19 + TypeScript	Component-based single-page user interface
Build tool	Vite 8	Fast development server and optimized production build
Styling	Tailwind CSS 4	Consistent utility-first responsive design
UI components	shadcn-style components, lucide-react	Buttons, inputs and icons
Routing	React Router 7	Role-based client-side routing and protected routes
Charts	Recharts	Analytics charts on coordinator and admin dashboards
Back-end / BaaS	Supabase	PostgreSQL database, authentication and REST API
Security	Row Level Security (RLS)	Row-wise data access control per user role
Version control	Git and GitHub	Collaborative development and change tracking
Deployment	Vercel	Hosting the web application with CI from GitHub
Linting	Oxlint	Static code analysis and code quality

Table 3.1: Tools and Technologies Used

3.5 Module Description

Module 1 - Authentication and Role Management: Handles registration and login using college e-mail, maintains user profiles and enforces the student, coordinator and admin roles with protected routes.

Module 2 - Club Discovery and Membership: Lists all clubs with search and category filters, shows club profiles, and manages the membership workflow (apply, approve, reject) with live status tracking.

Module 3 - Event Management and Registration: Allows coordinators to create events with venue, time and capacity; students can browse the event hub and register with a single tap.

Module 4 - Attendance Tracking: Enables quick check-in and marking of attendance for registered participants and maintains a complete attendance history per student.

Module 5 - Announcements: Provides a unified feed of club announcements so that important updates reach all members instantly.

Module 6 - Dashboards and Analytics: Presents role-based dashboards - student activity overview, coordinator club analytics and admin institution-level analytics using charts and reports.

Module 7 - Landing and Public Pages: An animated public landing page and public club directory to attract students and introduce the platform.

Chapter 4: Implementation and Results

4.1 Implementation Details

The application follows a modular folder structure: `src/pages` contains the views for each role, `src/components` contains reusable UI components and route guards, `src/contexts` provides the authentication context, `src/services` contains the data-access functions that call Supabase, and `src/types` holds the TypeScript interfaces for every database entity. A typical data-access service for fetching clubs is shown below:

```
export async function getClubs(): Promise<Club[]> {
  const { data, error } = await supabase
    .from("clubs").select("*").eq("status", "approved");
  if (error) throw error;
  return data ?? [];
}
```

The membership workflow is implemented as follows: a student submits an application which creates a record with status *pending*; the club coordinator reviews the application and updates the status to *approved* or *rejected*; the student dashboard reflects the change immediately. Event registration creates a record in the event_registrations table and prevents duplicate registrations. Attendance marking inserts or updates the attendance table for each participant. All data access is protected by Row Level Security policies so that students can view only their own records and coordinators can manage only their own club. Role-based routing is enforced using a protected route component that verifies the authenticated user's role before rendering a page.

4.2 Screenshots of the Application

The following figures present the key screens of the CCMS web application.

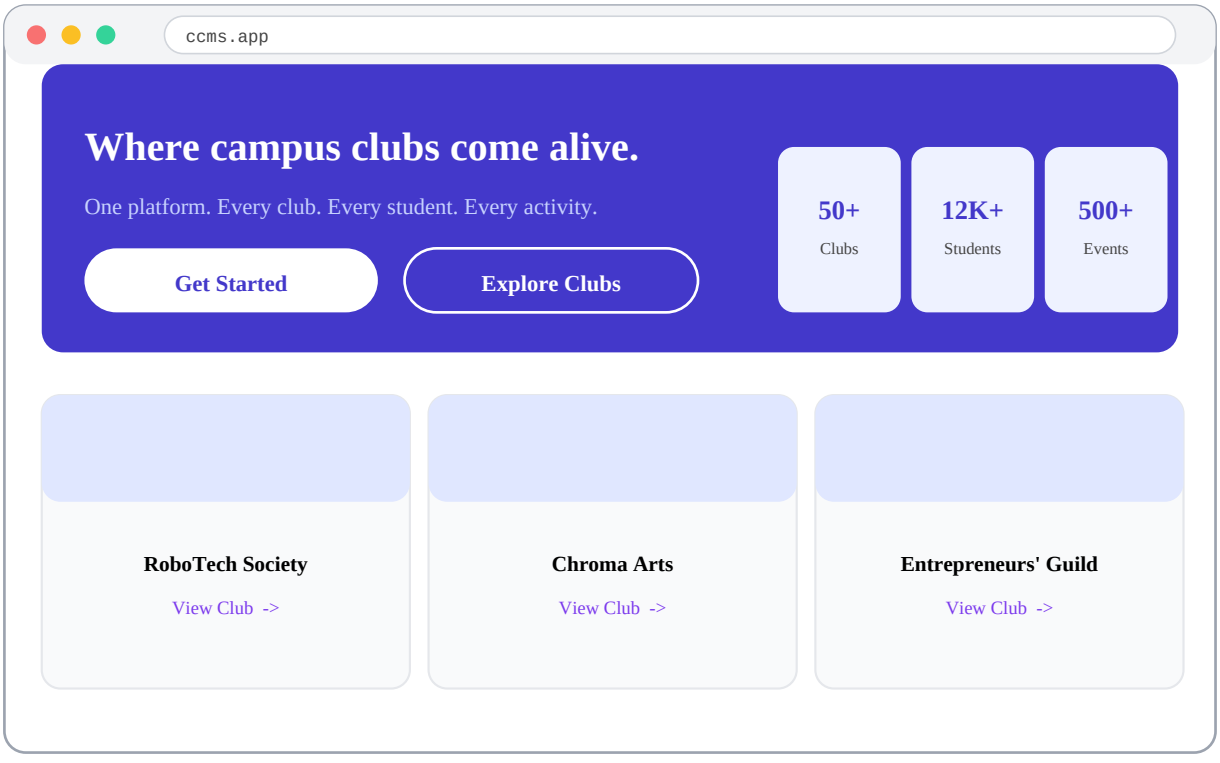


Figure 4.1: CCMS Landing Page

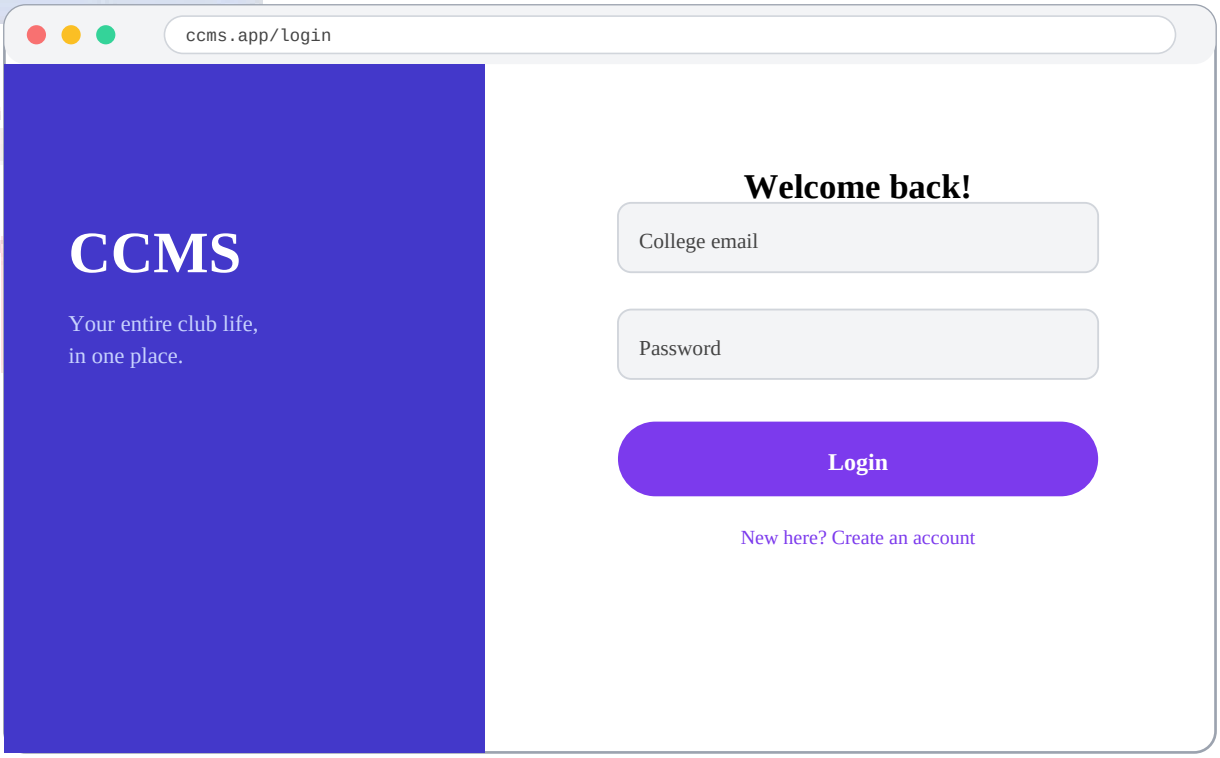


Figure 4.2: Login Page

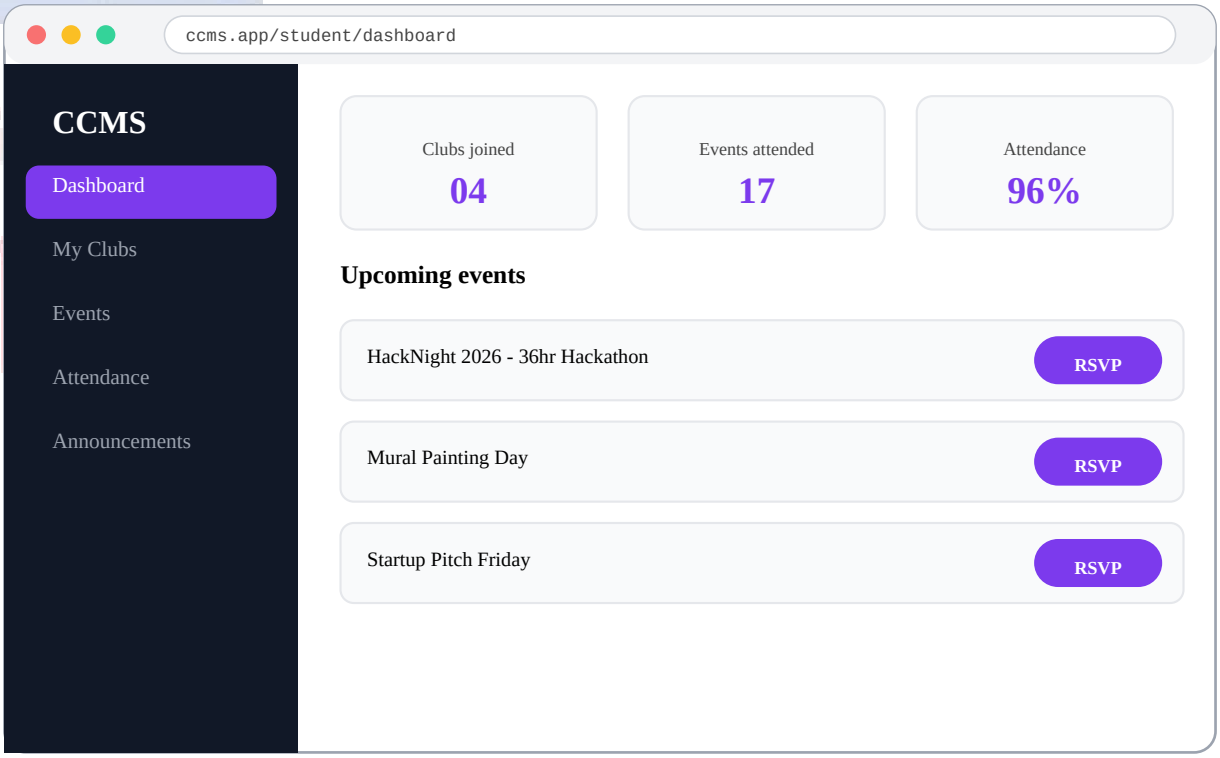


Figure 4.3: Student Dashboard

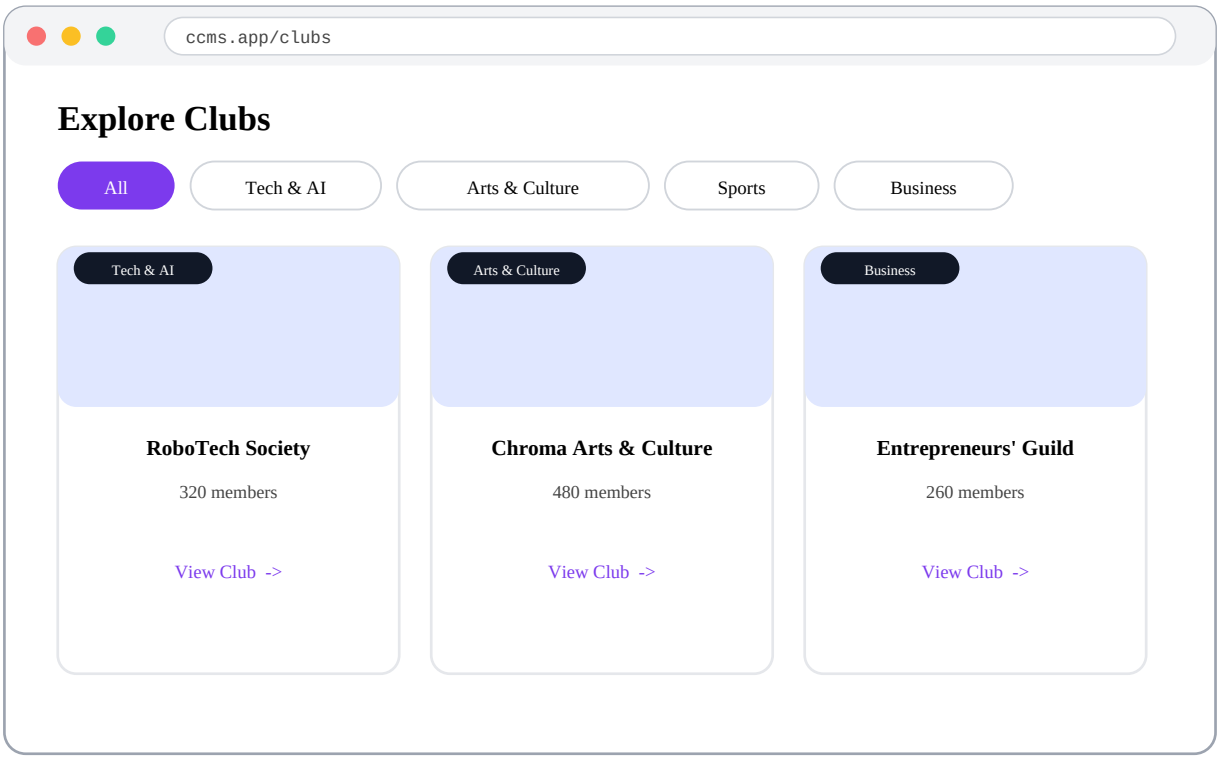


Figure 4.4: Club Discovery Page

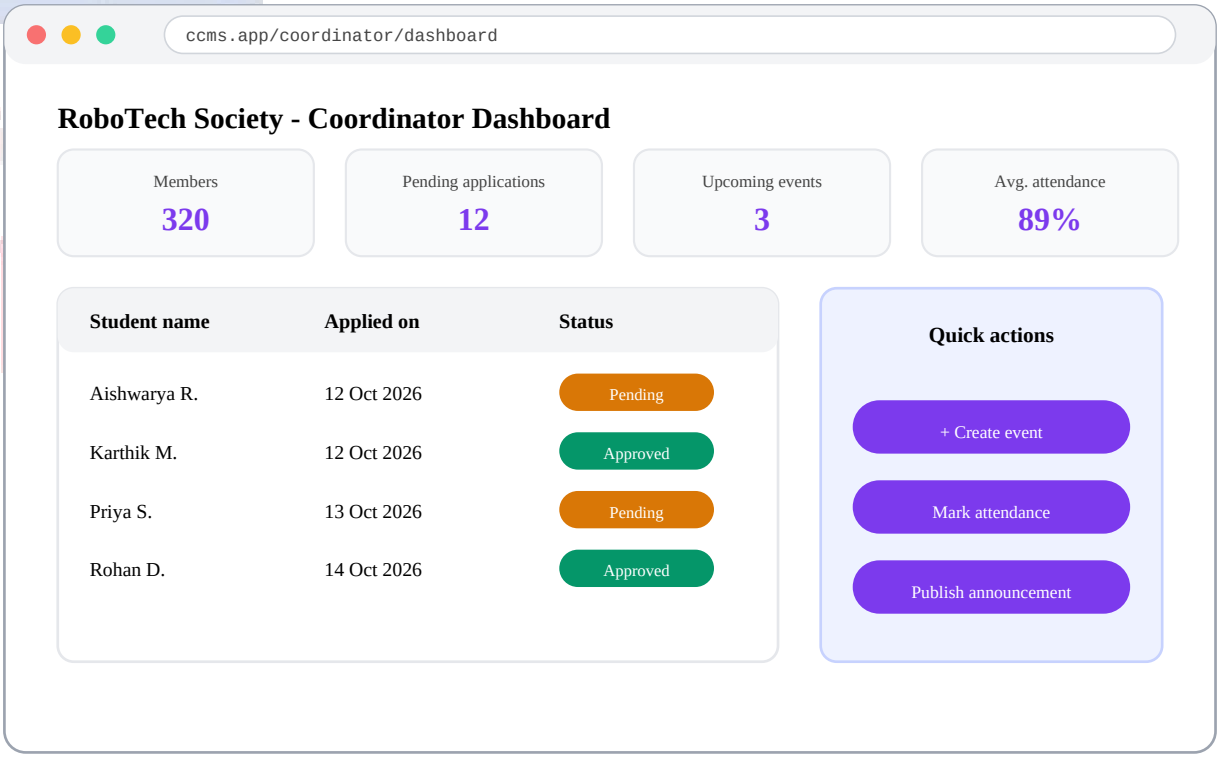


Figure 4.5: Coordinator Dashboard

4.3 Sample Inputs and Outputs

S.No	Sample Input / User Action	Output
1	Student logs in with valid college e-mail and password	Student dashboard is displayed with club and event summary
2	Student searches "robotics" in the club search box	RoboTech Society card is displayed with club details
3	Student clicks "Apply" on a club	Membership record created with status "pending"
4	Coordinator approves the pending membership	Status changes to "approved"; student sees the club under "My Clubs"
5	Student registers for "HackNight 2026" event	Registration confirmed; event appears in the student's calendar
6	Coordinator marks attendance for event participants	Attendance saved; student attendance history and percentage updated
7	Coordinator publishes an announcement	Announcement appears instantly in the feed of all club members

Table 4.2: Sample Inputs and Outputs

4.4 Unit Testing and Results

Unit testing was performed for the critical modules of the system. Each test case was executed with sample data and the actual result was verified against the expected result. The summary of the test cases is given in Table 4.1. All the test cases passed, and the system was found to behave as expected.

Test ID	Module	Test Scenario	Expected Result	Status

TC01	Authentication	Login with valid credentials	Dashboard is displayed	Pass
TC02	Authentication	Login with invalid password	Error message; login denied	Pass
TC03	Club Discovery	Search clubs by keyword and category	Matching clubs are filtered	Pass
TC04	Membership	Apply for club membership	Record created with status pending	Pass
TC05	Membership	Coordinator approves an application	Status changes to approved	Pass
TC06	Events	Register for an event within capacity	Registration succeeds	Pass
TC07	Events	Duplicate event registration	Duplicate registration is prevented	Pass
TC08	Attendance	Mark attendance for a participant	Attendance history is updated	Pass
TC09	Announcements	Publish a new announcement	Visible in member feed instantly	Pass
TC10	Security (RLS)	Access another club's data as coordinator	Access is denied by policy	Pass

Table 4.1: Unit Test Cases and Results

Chapter 5: Conclusion and Future Enhancement

5.1 Summary of the Work

In this micro project, the College Club Management System (CCMS) was designed, developed and tested as a centralized web platform for managing all club activities of the college. The complete software engineering process was followed - from requirement analysis and UML-based design to iterative implementation, unit testing and deployment. The system delivers club discovery, membership management, event registration, attendance tracking, announcements and role-based dashboards for students, club coordinators and administrators. The application is built with React, TypeScript and Tailwind CSS on the front end and Supabase (PostgreSQL, Authentication, Row Level Security) on the back end.

5.2 Conclusion

CCMS successfully replaces the fragmented workflow of Google Forms, QR codes, WhatsApp groups and spreadsheets with a single, secure and user-friendly platform. Students can now discover and join clubs, register for events and view their participation history in one place. Coordinators are able to manage members, events, attendance and announcements with significantly less manual effort. Administrators get a clear institution-level view of club activities. The systematic application of software engineering principles - modular design, separation of concerns, automated testing and version control - resulted in a maintainable and extensible system that meets the defined objectives.

5.3 Future Enhancements

- AI-powered club and event recommendations based on student interests and participation patterns.
- Native mobile application (React Native) with push notifications for announcements and events.
- QR-code based self check-in for event attendance.
- Automatic generation and verification of participation certificates.
- Online payment integration for paid events and club membership fees.
- Advanced analytics with machine learning insights on student engagement.
- Multilingual support and accessibility improvements for a wider student audience.

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PPT HANDOUTS
(4 Slides per page)

College Club Management System (CCMS)

- Micro Project - 2026-27 (Odd Semester)
- Course: 25CS312 Software Engineering
- Team: Naveen Kumar R V (URK2CS1016)
- Jeril Dany (URK25CS1)

Problem Statement

- Club info spread across forms, QR codes,
- WhatsApp groups and posters
- No unified membership / attendance record
- Manual effort for coordinators and admins

Objectives

- One platform for every club and event
- One-tap membership and event RSVP
- Automated attendance and portfolio
- Role-based dashboards and analytics

System Architecture

- Client: React + TypeScript SPA
- Application: Supabase API, Auth, RLS
- Data: PostgreSQL (8 core tables)
- Deployed on Vercel via GitHub

PPT HANDOUTS
(4 Slides per page)

Modules

- Auth & Role Management
- Club Discovery & Membership
- Events & Attendance
- Announcements, Dashboards & Analytics

Technology Stack

- React 19, TypeScript, Vite 8
- Tailwind CSS 4, lucide-react, Recharts
- Supabase (PostgreSQL + Auth + RLS)
- Git, GitHub, Vercel, Oxlint

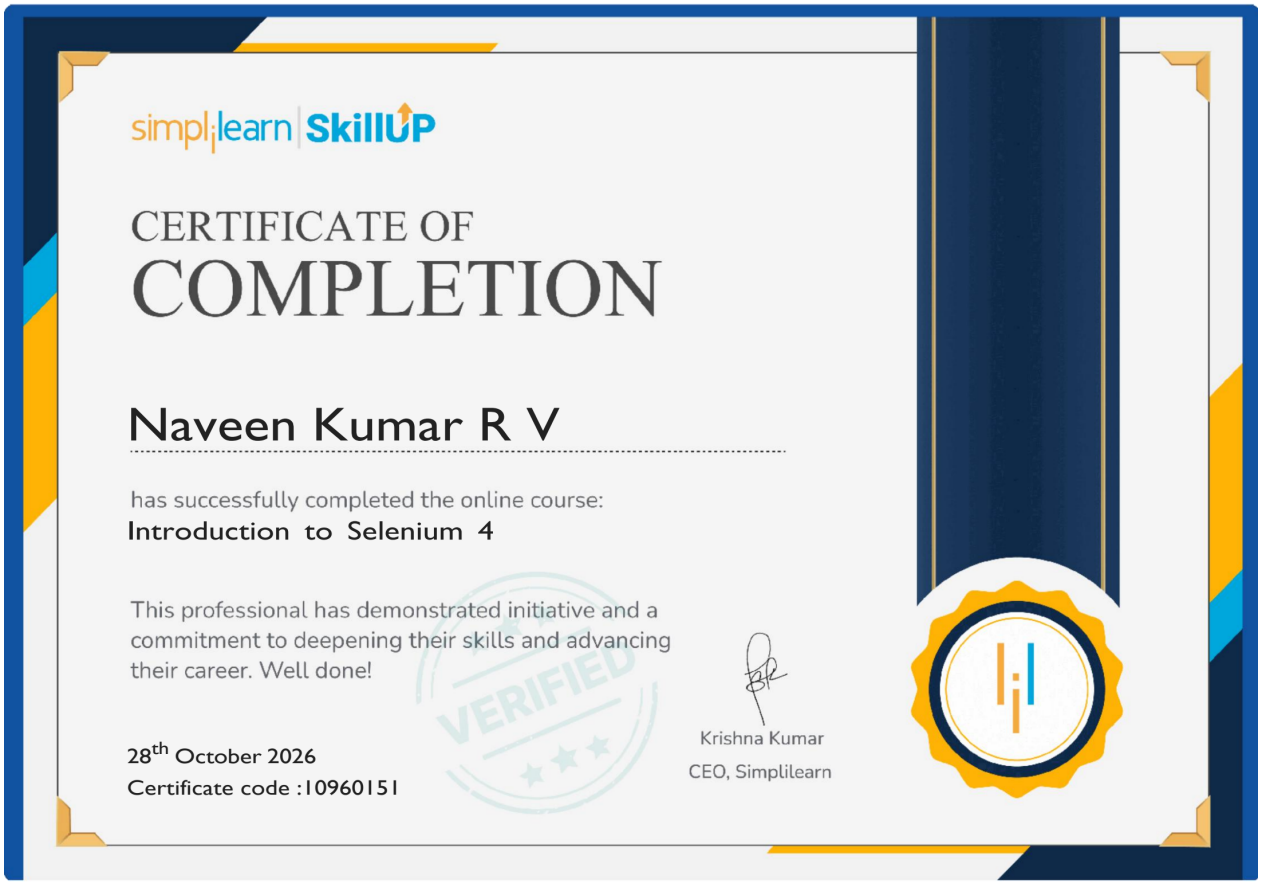
Implementation & Testing

- Service-based data access layer
- Protected role-based routing
- 10 unit test cases - all passed
- RLS policies secure every record

Conclusion & Future Scope

- Replaces scattered forms and chats
- Less manual effort; more participation
- Future: AI club recommendations
- Future: Mobile app, QR check-in, certificates

Introduction to Selenium 4 Certificate



EVALUATION SHEET

Reg. No. : URK2CS1016, URK25CS1

Name : Naveen Kumar R V, Jeril Dany

Course code : 25CS312

Course Name : Software Engineering

S.No	Rubrics	Maximum Marks	Marks Obtained
1	Problem Definition	6	
2	Application of SE concepts & Design	6	
3	Implementation	6	
4	Testing & Results	6	
5	Report & Viva	6	
6	Certificate Completion	10	
	Total	40	

Signature of the Faculty-in-charge